Unfolding syntax through prosody:
Signaling and interpreting branching structure in a tone language

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Introduction: Human sentence processing requires building hierarchical structure from a linear input stream, creating pressure to form syntactic groupings as early as possible [1]. Yet many structures demand that these early tendencies be delayed or revised, revealing a fundamental tension between incremental processing and the structural demands of the intended interpretation. This tension is a major source of structural ambiguity. For example, the phrase "*pink star hat"* allows both a left-branching (LB) interpretation (a hat with pink stars) or a right-branching (RB) interpretation (a pink hat with stars), see Fig. 1, left panel. Prosody provides a crucial mechanism for managing these competing structural possibilities. By marking boundaries through timing adjustments or pitch resets, language users can modulate the timing of constituent formation and guide interpretation toward the intended parse [2-3]. This study investigates how such cues support the incremental processing and production of LB/RB structures in Mandarin Chinese—a tone language where pitch is phonemic and prosodic disambiguation is correspondingly constrained. We address two questions central to the prosody–syntax interface: (i) How do tone speakers signal branching direction prosodically under tonal constraints? and (ii) How do they integrate different prosodic cues to recover the intended branching structure?

Methods: Two studies were conducted. The production study (**Exp 1**, N=16) used a sentence reading-out task with an audience design [4-5] to collect how Mandarin speakers use prosody to disambiguate the phrase, where branching type (LB vs. RB) and Contrastiveness (No Contrast vs. Contrast, depending on whether the visual scene included only LB or RB images or both) were manipulated (Fig. 1, middle panel). The analysis focuses on acoustic correlates—primarily timing and pitch variation—that mark branching distinctions. The perception study (**Exp 2**, N=32) manipulated branching type and prosodic cue (pause vs. lengthening vs. pause + lengthening) in a visual-world paradigm to examine how listeners integrate these cues during real-time syntactic interpretation (Fig. 1, right panel). In this design, LB cues were placed between *N1* and *N2*, whereas RB cues occurred between the *Adj* and *N1*. Participants completed a picture-identification task to find the referent intended by the auditory phrases, while their eye movements were recorded. This experiment includes three consecutive sessions, separated by brief breaks, to observe how exposure to different prosodic cues may lead to adaptive parsing behavior over time. Analyses focus on participants' picture choices and eye-movement patterns across cue conditions.

Results & Discussion: Both studies reveal a robust preference for LB structure, consistent with the alignment of left-to-right input and bottom-up structure building. This preference is evidenced by more frequent production of prosodic marking for LB when contrast is absent (LB pause 7.8 % vs. RB pause 3.3%; longer LB pause duration) and earlier eye fixations and more final choices favoring LB in comprehension (LB 80% vs. RB 60% overall). In **Exp 1**, speakers inserted silent pauses and lengthened the pre-boundary syllable—rather than using pitch reset or intensity—as cues for branching direction, with the location of these cues aligning with expected structural boundaries (Fig. 2). Pitch changes across boundaries showed variation, but these patterns were shaped by tonal requirements, limiting their reliability as disambiguating cues. In **Exp 2**, both RB and LB cues facilitated parsing relative to baseline, and sensitivity to these cues increased over time (Fig. 3). For LB interpretations, pause and pause + lengthening cues were more effective than lengthening alone, whereas cue types showed more similar effects for the syntactically less preferred RB interpretation. Moreover, cue effectiveness also shifted across sessions; notably, RB pause elicited a higher proportion of RB interpretations in Session 2 (Fig. 4). These suggest that prosody functions as a dynamic probabilistic cue in syntactic parsing, with listeners adjust its weight based on their structural expectations and their experience with the cue-parse mapping. Together, these findings show that (i) tone language speakers mark branching direction through timing-based prosodic cues such as silent pauses aligned with structural boundaries, and (ii) they also use temporal boundary cues to guide structural interpretation and identify branching structures.

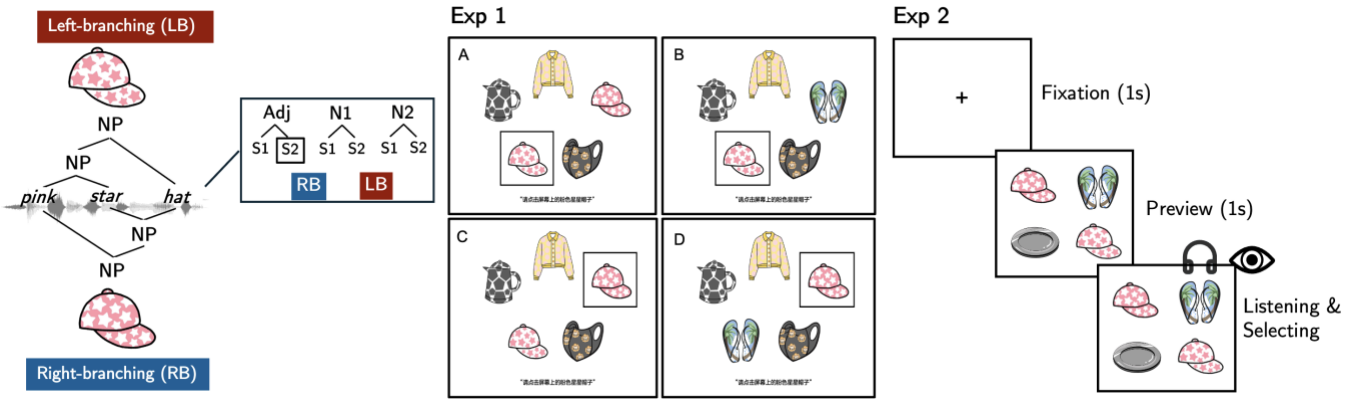
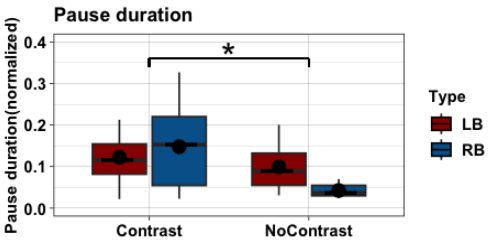
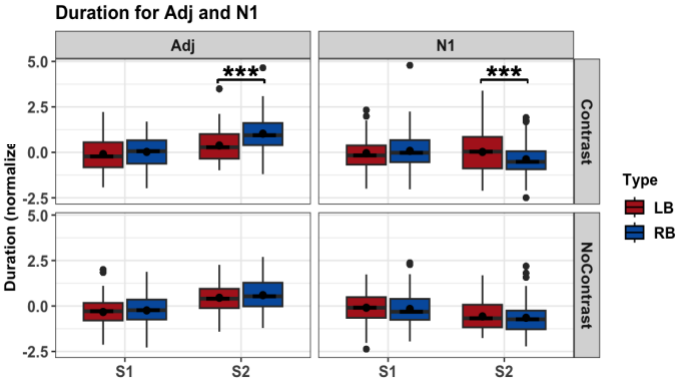


Fig. 1 **Left**: Mandarin branching structure with 3 components totaling 6 syllables. **Middle**: Exp 1: selected images are Contrast LB (A), NoContrast LB (B), Contrast RB (C), and NoContrast RB (D). Speakers read the sentence "*please click on the pink star hat*" aloud for hypothetical listeners completing a picture-recognition task. **Right**: Exp 2: listeners heard the phrases and chose the matching pictures

Fig. 2 **Left**: Pause duration. **Right**: Pre-boundary syllable lengthening in Exp 1. S1: first syllable, S2: second syllable





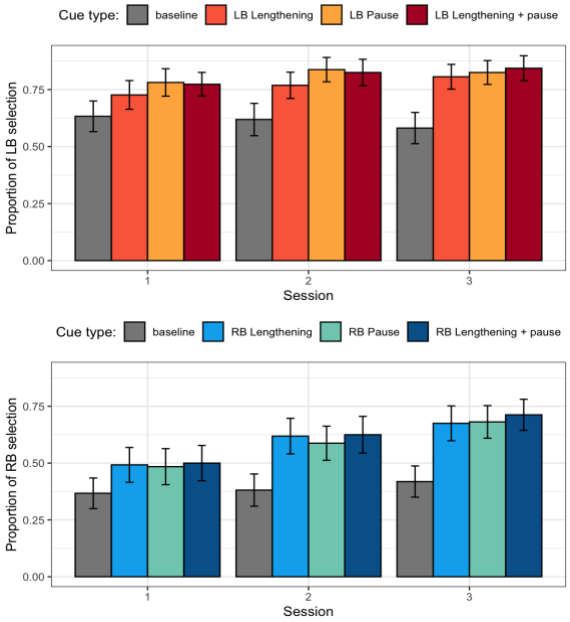


Fig. 3 LB & RB choices across cues and sessions in Exp 2

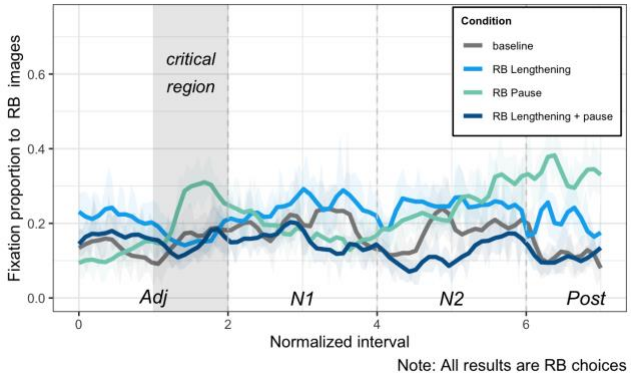


Fig. 4 Fixation proportion to RB images in Session 2 in Exp 2

References: [1] Frazier, 1987 . [2] Snedeker & Trueswell, 2003. [3] Watson & Gibson, 2004. [4] Clark & Murphy, 1982. [5] Lockridge & Brennan, 2002.